Weixuan Li

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Education

Johns Hopkins University (JHU), USA **Sept. 2023 – Present**
M.S. in Engineering (Expected), Dept. of Mechanical Engineering
- Cumulative GPA: 4.0/4.0; TOEFL: 113/120

Massachusetts Institute of Technology (MIT), USA **Sept. 2022 – May 2023**
MIT-SUSTech special student program (Non-degree), Dept. of Mechanical Engineering
- Cumulative GPA: 5.0/5.0

Southern University of Science and Technology (SUSTech), China **Sept. 2019 – June 2023**
B.S. in Theoretical and Applied Mechanics, Dept. of Mechanical and Aerospace Engineering
- Cumulative GPA: 3.86/4.0; Class Rank: 1/20

Publications

[1] **W. Li**, S. Kuhar, J.H. Seo, R. Mittal. (2024). Modeling the Effect of Sleeve Gastrectomy on Gastric Digestion in the Stomach: Insights from Multiphase Flow Modeling. arXiv preprint arXiv:2411.18586. Submitted to *Journal of Biomechanical Engineering* (Under review).

[2] Krinos, A.I., Shapiro, S.K., **Li, W.**, Haley, S.T., Dyhrman, S.T., Dutkiewicz, S., Follows, M.J., and Alexander, H. (2024). Intraspecific diversity in thermal performance determines phytoplankton ecological niche. bioRxiv. DOI: 10.1101/2024.02.14.580366. Submitted to *Ecology Letters* (Under review).

Conference presentations

[1] **Li, W.**, Kuhar, S., Seo, J. H., Mittal, R. (2024). Food Digestion in the Stomach after Bariatric Surgery: Insights from Multiphase Flow Modeling. *Bulletin of the American Physical Society*.

Research Experience

Computational Modeling of Stomach Surgery Using Multiphase Flow Simulations **Sept. 2023 - Present**
Flow Physics and Computation Lab, supervised by Prof. ***Rajat Mittal*** *JHU, Baltimore, MD, USA*

Project 1: Modeling the Effect of Sleeve Gastrectomy on Gastric Digestion in the Stomach: Insights from Multiphase Flow Modeling **Jan. 2024 - Present**
- Presented at the *APS Division of Fluid Dynamics* (2024), showcasing insights into post-surgery gastric dynamics through multiphase flow modeling.
- Developed multiple post-operative stomach models with varying resection volumes and motility patterns to explore the impact of these changes on gastric emptying and mixing efficiency.
- Conducted multiphase flow simulations and compared results between pre- and post-surgery stomach models.
- Investigated the relationship between sleeve size, intragastric pressure, and gastric emptying rate, finding that reduced sleeve size leads to increased intragastric pressure, resulting in accelerated emptying.
- Quantified the impact of altered gastric motility on mixing rates, showing that reduced motility impairs mixing efficiency, which subsequently slows the post-surgery emptying.

Project 2: ViCar3D Code Validation with Multiphase Flow Simulations **Sept. 2023 - Dec. 2023**
- Conducted literature review and selected a relevant study titled *"Simulation of the Falling Droplet by the Lattice Boltzmann Method"* to benchmark ViCar3D, a custom multiphase flow simulation code developed within our research group.
- Reproduced 2D falling droplet simulations from the study; results from ViCar3D aligned well with the literature, validating the code's accuracy and reliability.

Computational Modeling of Marine Ecosystems: Phytoplankton Thermal Diversity Research **Sept. 2022 - May 2023**
Mick Follow's Group, supervised by Prof. ***Mick Follows*** *MIT, Cambridge, MA, USA*
- Co-authored a manuscript submitted to *Ecology Letters*, currently under peer review, where I contributed as the third author, focusing on the thermal niche modeling of marine phytoplankton.
- Conducted literature reviews on marine microbial ecology and biogeochemical cycles, with an emphasis on the interactive effects of environmental factors, particularly temperature, on the dynamics of marine phytoplankton such as coccolithophores.
- Analyzed Darwin ecosystem model outputs to assess and visualize the thermal range variations of different *Gephyrocapsa huxleyi* strains across global ocean regions, identifying patterns linked to climate variability.

- Contributed to the development of Figure 4 (panels B-F) in the manuscript, which involved probabilistic modeling and projection of the environmental distributions of *Gephyrocapsa huxleyi* strains using thermal response data and statistical tools.

Large-eddy Simulation (LES) in Lattice Boltzmann Method for Turbulent Channel Flow **Sept. 2021 - Aug. 2022**
Wang's Research Group, supervised by Prof. ***Lian-Ping Wang*** *SUSTech, Shenzhen, China*

- Conducted advanced theoretical analysis and applications of LES-LBM for turbulent flows, with a focus on insights from "Large Eddy Simulation for Incompressible Flows" by Pierre Sagaut.
- Developed and implemented FORTRAN90 codes to integrate the Smagorinsky subgrid-scale model and the Musker wall model into a custom numerical simulation package using Linux, enhancing its efficiency while maintaining accuracy in simulating turbulent channel flows.
- Awarded the Excellent Undergraduate Thesis Award at SUSTech.
- Secured funding of RMB 10,000 ($1,500) for this project from the National Innovation and Entrepreneurship Training Program for College Students.

Awards and Fellowships

Johns Hopkins Mechanical Engineering Distinguished Master's Fellowship **Aug. 2023**

- Selected as one of the most excellent master's students to receive the fellowship (only 3-4 students each year), which provides 50% tuition coverage during the master's program.

Thomas Sheridan Award | *MIT Department of Mechanical Engineering Student Award* **May 2023**

- Distinguished as one of a select group of undergraduate and graduate students recognized for outstanding achievements.

Excellent Graduate for Exceptional Performance **June 2023**

- Named Excellent Graduate of Southern University of Science and Technology.

Excellent Undergraduate Thesis Award **June 2023**

- Received Outstanding Undergraduate Thesis Award for "Large-eddy Simulation with Wall Model in Lattice Boltzmann Method for Turbulent Channel Flow" at Southern University of Science and Technology.

ASC22 Student Supercomputer Challenge | *Second Prize* **May 2022**

- Won Second Prize in ASC22 Student Supercomputer Challenge, one of the world's major student supercomputing competitions, as a member of the Southern University of Science and Technology team.

Merit Student Scholarship | *First Class* **Nov. 2020, 2021, 2022**

- Awarded to top 5% of undergraduate students for academic excellence.

Work Experience

Undergraduate Research Mentor | *Johns Hopkins University* **Oct. 2024 – Present**

- Mentored undergraduate student *Ceyda Turna* in computational research on high-performance computing systems (*Rockfish*), training her in modifying code, running simulations, and post-processing data.

Teaching Assistant | *Statistical Learning for Engineers, Johns Hopkins University* **Sept. 2024 – Present**

- Selected as a teaching assistant for "Statistical Learning for Engineers" based on outstanding performance in this course.
- Assisted 40 graduate students. Responsibilities include grading assignments and exams, holding office hours, managing online platform *Piazza*, and conducting lectures during the instructor's travels.

Student Assistant | *Southern University of Science and Technology* **Sept. 2019 – May 2020**

- Organized meetings for tutor group leaders, managed college branding activities, and revised student manuals.

leadership and Activities

Secretary-General of the Student Union | *Shuren College, SUSTech* **Sept. 2020 – Sept. 2021**

- Organized events, drafted documents, managed files, and coordinated with other organizations.

Vice President of the Table Tennis Club | *SUSTech* **Sept. 2020 – Sept. 2021**

- Authored publicity articles, supported the President in organizing events, and contributed to the club's recognition as one of the annual top 10 clubs of the university.

Volunteer Instructor, STEAM+ Team | *SUSTech* **Sept. 2019 – Dec. 2019**

- Taught primary students in the community to use 3D printers through volunteer classes.
- Contributed to the project being recognized as the "Annual Outstanding Public Welfare Project" by the SUSTech Volunteer Association.

Technical Skills

Programming Languages: Fortran90, Python, Java, Julia
Software and Tools: Paraview, Tecplot, MATLAB, COMSOL, ANASYS, Solidworks, Blender, Linux, High-Performance Computing (HPC) systems